

Aufgabe 2

Zusatz zur Lösung: $f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$

$$\begin{aligned} f_{X,Y}(x, y) &= \left(\frac{1}{\sqrt{2\pi}} \right)^2 \frac{1}{\sqrt{\det(\Sigma)}} \exp \left(-\frac{\left(\begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \right)^\top \Sigma^{-1} \left(\begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \right)}{2} \right) \\ &= \left(\frac{1}{\sqrt{2\pi}} \right)^2 \frac{1}{\sqrt{\sigma_{11}\sigma_{22}}} \exp \left(-\frac{\begin{pmatrix} x-\mu_1 \\ y-\mu_2 \end{pmatrix}^\top \begin{pmatrix} \sigma_{11}^{-1} & 0 \\ 0 & \sigma_{22}^{-1} \end{pmatrix} \begin{pmatrix} x-\mu_1 \\ y-\mu_2 \end{pmatrix}}{2} \right) \\ &= \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{\sigma_{11}}} \cdot \frac{1}{\sqrt{\sigma_{22}}} \exp \left(-\frac{(x-\mu_1)^2 \sigma_{11}^{-1} + (y-\mu_2)^2 \sigma_{22}^{-1}}{2} \right) \\ &= \frac{1}{\sqrt{2\pi\sigma_{11}}} \exp \left(-\frac{1}{2\sigma_{11}} (x-\mu_1)^2 \right) \cdot \frac{1}{\sqrt{2\pi\sigma_{22}}} \exp \left(-\frac{1}{2\sigma_{22}} (y-\mu_2)^2 \right) \\ &= f_X(x) \cdot f_Y(y) \end{aligned}$$